Umar Jahangir

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Professional Summary

Computer Science and Engineering (IoT, Cybersecurity & Blockchain Technology) student building end-to-end full-stack and AI-driven applications, with hands-on experience across mobile development, cybersecurity, and applied ML/embedded systems. Holds 7 patents and 1 research paper across decentralized, healthcare, AI, retail analytics, and IoT domains.

Education

Vishwakarma Institute of Technology, Pune *Aug 2024 – 2028 (Present)*

B.Tech in Computer Science and Engineering (IoT, Cybersecurity & Blockchain Technology)

CGPA: 9.09 / 10.0 **Patents**: 7 **Research Paper**: 1

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, C, Solidity

Frontend: Expo, React, Next.js, HTML, CSS, Tailwind CSS, Bootstrap, jQuery

Backend: Node.js, Express.js, Docker, Flask, FastAPI, REST APIs, JWT Auth

AI/ML & Data: TensorFlow, OpenCV, Mediapipe, SpaCy, NumPy, Pandas, Matplotlib, Gemini API, AssemblyAI

Databases: MySQL, MongoDB, PostgreSQL

Security & Tools: Kali Linux, Wireshark, Nmap, John The Ripper, Hydra, hashcat, OWASP ZAP, GPG

Other: Linux, Streamlit, Git, GitHub, Arduino, Raspberry Pi, Figma, Power BI, Looker, Excel, Cisco Packet Tracer, WEKA

Professional Experience

Cybersecurity Intern — Algo Analytics Pvt. Ltd. — Pune, India *Dec 2025 – Mar 2026*

- Deployed and configured the **Aksha AI-powered surveillance platform** using Docker/containerized services to create controlled environments for security testing.
- Performed authorized penetration testing and vulnerability assessment using **Kali Linux**, conducting network reconnaissance and service enumeration with **Nmap** and traffic analysis with **Wireshark** to identify exposed services and potential attack surfaces.
- Tested application security against **SQL Injection, NoSQL Injection**, and **JWT**-based authentication/authorization weaknesses across API endpoints and data-access components.

Full Stack Intern — Algo Analytics Pvt. Ltd. — Pune, India *Jul 2025 – Nov 2025*

React Native, Expo, Axios, REST APIs, AWS S3, Node.js/Express, Redux, JWT

- Developed a **React Native/Expo invoice management app** with camera capture, QR scanning, gallery upload, image processing, and AWS S3 integration.
- Implemented **role-based authentication and approval workflows** using JWT, Redux, Context API, and Expo Router.
- Built and integrated **REST API services** with Axios, QR data parsing, error handling, and debugging; resolved authentication and API integration issues across frontend and backend. .

Software Development Engineer Intern (Industrial Project)—Yantriki Soft — Pune, India *Feb 2025 – Jun 2025*

Retail Causal AI – Multi-Stakeholder GNN-Causal-RL Framework for Demand Forecasting & Inventory Optimization

- Built a retail analytics platform combining **demand forecasting, GNN-based product modeling, and RL-driven inventory optimization**.
- Implemented **PPO reinforcement learning** and causal analysis using **Propensity Score Matching, DiD, and Double ML**, achieving **MAE of 5** on real-world retail data.
- Added **SHAP explainability and PSI-based drift monitoring**; patent application in preparation.

Projects

CREDIT – AI-Assisted Alternative Credit Underwriting — *Top 10 Finalist, MIT ADT AI Grand Challenge*

React, TypeScript, Node.js, MongoDB, Python, XGBoost, Random Forest, Scikit-learn, Pandas, NumPy

- Built a digital loan-underwriting platform combining **rule-based screening, ML risk scoring, XGBoost, explainable AI, and AWS Textract OCR** for thin-file borrower assessment.
- Integrated the **Python/XGBoost risk service with a Node.js backend** and developed an optional **Ollama-hosted LLM copilot** for analyst Q&A and review support .

InnoVelty AI – Idea Novelty Analyzer — LLM, NLP, Semantic Search

- AI tool that evaluates idea novelty against patents and research papers using LLMs, with semantic search over vector embeddings for prior-art retrieval and LLM-based novelty scoring.
- Surfaces the top 10 matched patents and research papers with source links to cut down manual reference searching, plus an AI chatbot that suggests ways to enhance the user's idea.

FairTrust AI – Bias Auditing for Hiring Algorithms — *Team Project, PCCOE Hackathon 4.0*

Python, Flask, SHAP, Fairlearn, XGBoost

- Built a hackathon auditing framework evaluating hiring models for bias across 7 fairness dimensions (demographic parity, calibration, intersectional bias, etc.) before deployment.
- Produced a weighted Ethical Score and role-based compliance reports for developers, regulators, and candidates, using SHAP-based explainability for EU AI Act / GDPR-ready certification reports.

Achievements

- Internal SIH 2026 **Winner** – Vishwakarma Institute of Technology
- **Top 10 Finalist** – MIT(ADT) AI Grand Challenge (CREDIT – AI-Assisted Alternative Credit Underwriting)
- Patents - **7** — Research paper - **1**